$$
\begin{aligned}
&\lambda_2 \to \frac{1}{4}\left(g_1^2+g_2^2\right)+\\
&\frac{1}{16\pi^2}\Big(\frac{3}{8}g_1^4+g_1^2g_2^2+\frac{23}{24}g_2^4+\frac{4}{9}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,0}[m_2]+\frac{2}{3}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{4,-1}[m_2]-\frac{32}{45}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{5,-2}[m_2]-\\
&\frac{1}{12}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde{d}}^{p}]+g_1^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{2,0}[m_{\tilde{d}}^{r}]-\frac{1}{4}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde{e}}^{p}]+\\
&g_1^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{2,0}[m_{\tilde{e}}^{r}]+\frac{1}{8}\left(4\,\overline{y_e}^{pr}y_e^{pr}(-g_1^2+g_2^2)-\sum_p(g_1^4+g_2^4)\right)\mathrm{LF}_{2,0}[m_{\tilde{l}}^{p}]+\frac{1}{18}C_{\phi 2^2}\\
&\quad g_2^2\left(36\,\overline{y_e}^{pr}y_e^{pr}-5\sum_p g_2^2\right)\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]+\frac{1}{12}C_{\phi 2^2}g_2^2\left(-24\,\overline{y_e}^{pr}y_e^{pr}+\sum_p g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]+\\
&\frac{8}{45}\sum_p C_{\phi 2^2}g_2^4\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]+\left(\frac{1}{2}\overline{y_d}^{pr}y_d^{pr}(g_1^2+3g_2^2)-\frac{1}{24}\sum_p(g_1^4+9g_2^4)\right)\mathrm{LF}_{2,0}[m_{\tilde{q}}^{p}]+\\
&\frac{1}{6}C_{\phi 2^2}g_2^2\left(36\,\overline{y_d}^{pr}y_d^{pr}-5\sum_p g_2^2\right)\mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]+\frac{1}{4}C_{\phi 2^2}g_2^2\left(-24\,\overline{y_d}^{pr}y_d^{pr}+\sum_p g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]+\\
&\frac{8}{15}\sum_p C_{\phi 2^2}g_2^4\,\mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]-\frac{1}{3}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde{u}}^{p}]+\frac{2}{9}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]+\\
&\frac{1}{3}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]-\frac{16}{45}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]+\frac{1}{2}g_1^4\,\mathrm{LF}_{2,2,-2}[m_1,\tilde{\mu}]+\\
&\frac{1}{2}g_1^4m_1^2\,\mathrm{LF}_{2,2,-1}[m_1,\tilde{\mu}]-\frac{8}{3}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{2,1,0}[m_2,\tilde{\mu}]+\frac{5}{2}g_2^4\,\mathrm{LF}_{2,2,-2}[m_2,\tilde{\mu}]+\\
&\frac{1}{6}g_2^4(-34C_{\phi 2^2}+3m_2^2)\,\mathrm{LF}_{2,2,-1}[m_2,\tilde{\mu}]+\frac{2}{3}m_2\tilde{\mu}g_2^4(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\,\mathrm{LF}_{2,2,0}[m_2,\tilde{\mu}]+\\
&\frac{4}{3}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,1,-1}[m_2,\tilde{\mu}]+\frac{28}{3}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,2,-2}[m_2,\tilde{\mu}]-\\
&\frac{1}{3}m_2g_2^4\left(18m_2C_{\phi 2^2}+\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\right)\mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}]-4C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,3,-3}[m_2,\tilde{\mu}]+\\
&4C_{\phi 2^2}g_2^4m_2^2\,\mathrm{LF}_{3,3,-2}[m_2,\tilde{\mu}]-4C_{\phi 2^2}g_2^4\,\mathrm{LF}_{4,2,-3}[m_2,\tilde{\mu}]+4C_{\phi 2^2}g_2^4m_2^2\,\mathrm{LF}_{4,2,-2}[m_2,\tilde{\mu}]-\\
&3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+g_1^2\,\overline{a_d}^{pr}a_d^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+\\
&\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)\mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-\\
&\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+g_1^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+\\
&\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)\mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+\\
&g_2^2\,\overline{a_e}^{pr}a_e^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\frac{1}{2}\overline{a_e}^{pr}a_e^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\\
&g_2^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)\mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\\
&\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\\
&\frac{1}{4}\left(-2C_{\phi 2^2}\overline{a_e}^{pr}a_e^{pr}(3g_1^2+5g_2^2)-\tilde{\mu}(3g_1^2+7g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{pr}a_e^{pr}+C_{\phi 1\phi 2}y_e^{pr}\overline{a_e}^{pr})\right)\\
&\quad\mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\\
&\frac{1}{3}\left(2C_{\phi 2^2}\overline{a_e}^{pr}a_e^{pr}(3g_1^2+g_2^2)+\tilde{\mu}(3g_1^2+2g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{pr}a_e^{pr}+C_{\phi 1\phi 2}y_e^{pr}\overline{a_e}^{pr})\right)\\
&\quad\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]-\\
&2C_{\phi 2^2}\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+2C_{\phi 2^2}\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+\\
&\overline{a_d}^{pr}a_d^{pr}(2g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-\frac{3}{2}\overline{a_d}^{pr}a_d^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+\\
&(2g_1^2+3g_2^2)\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-\\
&\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+\\
&\frac{1}{4}\left(-6C_{\phi 2^2}\overline{a_d}^{pr}a_d^{pr}(7g_1^2+5g_2^2)-21\tilde{\mu}(g_1^2+g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{pr}a_d^{pr}+C_{\phi 1\phi 2}y_d^{pr}\overline{a_d}^{pr})\right)\\
&\quad\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+\\
&\left(2C_{\phi 2^2}\overline{a_d}^{pr}a_d^{pr}(3g_1^2+g_2^2)+\tilde{\mu}(3g_1^2+2g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{pr}a_d^{pr}+C_{\phi 1\phi 2}y_d^{pr}\overline{a_d}^{pr})\right)\\
&\quad\mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-\\
&6C_{\phi 2^2}\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+6C_{\phi 2^2}\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+\\
&\frac{1}{2}\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\\
&\frac{1}{4}\tilde{\mu}\left((g_1^2-g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr})+2C_{\phi 2^2}\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\right)\mathrm{LF}_{2,2,0}[\\
&\quad m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\frac{3}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr})\right)\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\\
&\frac{1}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr})\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\\
&\frac{3}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr})\right)\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\\
&\frac{1}{2}\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\frac{3}{2}\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+\\
&\left(-\frac{1}{2}\tilde{\mu}(g_1^2-2g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr})+C_{\phi 2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(-g_1^2+g_2^2)\right)\\
&\quad\mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+\frac{1}{4}\tilde{\mu}\Big(-(g_1^2-g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr})-\\
&\qquad 2C_{\phi 2^2}\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\Big)\mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+\\
&\left(-\frac{3}{2}\tilde{\mu}(g_1^2+2g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr})-3C_{\phi 2^2}\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\right)\\
&\quad\mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+\\
&\tilde{\mu}\left((3g_1^2+2g_2^2)(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr})+2C_{\phi 2^2}\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pr}(3g_1^2+g_2^2)\right)\\
&\quad\mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\frac{3}{4}g_1^2(g_1^2+g_2^2)\,\mathrm{LF}_{2,1,-1}[\tilde{\mu},m_1]+\frac{1}{2}g_1^2(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-2}[\tilde{\mu},m_1]-\\
&2C_{\phi 2^2}g_1^2(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_1]+\frac{1}{2}m_1\tilde{\mu}g_1^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,0}[\tilde{\mu},m_1]+\\
&\frac{3}{2}C_{\phi 2^2}g_1^4\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_1]+\frac{1}{2}m_1g_1^4\left(3m_1C_{\phi 2^2}-2\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\right)\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_1]+\\
&\frac{1}{3}C_{\phi 2^2}g_1^2(12g_1^2+7g_2^2)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]-\\
&\frac{1}{4}m_1\tilde{\mu}g_1^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(6g_1^2+5g_2^2)\,\mathrm{LF}_{4,1,-1}[\tilde{\mu},m_1]-C_{\phi 2^2}g_1^4\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_1]+\\
&m_1g_1^4\left(-m_1C_{\phi 2^2}+\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\right)\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_1]-\frac{2}{3}C_{\phi 2^2}g_1^2(3g_1^2+g_2^2)\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]+\\
&\frac{1}{3}m_1\tilde{\mu}g_1^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(3g_1^2+2g_2^2)\,\mathrm{LF}_{5,1,-2}[\tilde{\mu},m_1]-\frac{9}{4}g_2^2(g_1^2+g_2^2)\,\mathrm{LF}_{2,1,-1}[\tilde{\mu},m_2]+\\
&\frac{3}{2}g_2^2(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-2}[\tilde{\mu},m_2]-\frac{2}{3}C_{\phi 2^2}g_2^2(9g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_2]+\\
&\frac{1}{6}m_2\tilde{\mu}g_2^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(9g_1^2+5g_2^2)\,\mathrm{LF}_{3,1,0}[\tilde{\mu},m_2]+\frac{85}{6}C_{\phi 2^2}g_2^4\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_2]+\\
&\frac{1}{6}m_2g_2^4\left(9m_2C_{\phi 2^2}-28\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\right)\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_2]+\\
&3C_{\phi 2^2}g_2^2(4g_1^2+g_2^2)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]-\\
&\frac{1}{4}m_2\tilde{\mu}g_2^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(18g_1^2+11g_2^2)\,\mathrm{LF}_{4,1,-1}[\tilde{\mu},m_2]-7C_{\phi 2^2}g_2^4\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_2]+\\
&m_2g_2^4\left(-m_2C_{\phi 2^2}+4\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\right)\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_2]-2C_{\phi 2^2}g_2^2(3g_1^2+g_2^2)\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]+\\
&m_2\tilde{\mu}g_2^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(3g_1^2+2g_2^2)\,\mathrm{LF}_{5,1,-2}[\tilde{\mu},m_2]+8C_{\phi 2^2}g_1^2g_2^2\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_1]+\\
&\frac{1}{2}g_1^2g_2^2\left(12m_1m_2C_{\phi 2^2}-\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(7m_1+5m_2)\right)\mathrm{LF}_{2,2,1,-1}[m_2,\tilde{\mu},m_1]-\\
&4C_{\phi 2^2}g_1^2g_2^2\,\mathrm{LF}_{3,2,1,-3}[m_2,\tilde{\mu},m_1]+2g_1^2g_2^2\left(-2m_1m_2C_{\phi 2^2}+\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(m_1+m_2)\right)\\
&\quad\mathrm{LF}_{3,2,1,-2}[m_2,\tilde{\mu},m_1]-6\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}\left(-\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi 1\phi 2}y_d^{pt}\overline{a_d}^{pr}\right)\mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t}]-\\
&6\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{pr}a_d^{pt}-C_{\phi 1\phi 2}y_d^{pt}\overline{a_d}^{pr}\right)\mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-\\
&2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+\\
&\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}\left(-\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi 1\phi 2}y_e^{pt}\overline{a_e}^{pr}\right)\mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t}]-\\
&2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+\\
&\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{pr}a_e^{pt}-C_{\phi 1\phi 2}y_e^{pt}\overline{a_e}^{pr}\right)\mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\\
&\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pt})\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+\\
&\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pt})\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-\\
&2C_{\phi 2^2}\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+\\
&2C_{\phi 2^2}\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+\frac{1}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\overline{a_e}^{pr}\\
&\quad\mathrm{LF}_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r}]+\frac{1}{2}\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}]-\\
&2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}(C_{\phi 2^2}a_e^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{sr})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+\\
&2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}(C_{\phi 2^2}a_e^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{sr})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-\\
&2\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+\\
&2\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-\\
&3\,\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pt})\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+\\
&3\,\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pt})\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]-\\
&6C_{\phi 2^2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+6C_{\phi 2^2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\\
&\quad\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\overline{a_d}^{pr}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}]+\\
&\frac{3}{2}\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}]+\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\\
&\quad\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]-\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]-\\
&\frac{3}{4}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s}]-\\
&\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]-\\
&6\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}(C_{\phi 2^2}a_d^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{sr})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+\\
&6\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}(C_{\phi 2^2}a_d^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{sr})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-\\
&6\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&6\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\\
&\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-\\
&\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-\\
&\frac{3}{4}C_{\phi 1\phi 2}\tilde{\mu}\,\overline{y_d}^{st}y_d^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\\
&\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}]+\\
&g_1^2g_2^2\,\mathrm{LF}_{2,1,1,-2}[\tilde{\mu},m_1,m_2]+g_1^2g_2^2(-3C_{\phi 2^2}+m_1m_2)\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_1,m_2]+\\
&m_1\tilde{\mu}g_1^2g_2^2(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_1,m_2]+7C_{\phi 2^2}g_1^2g_2^2\,\mathrm{LF}_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+\\
&\frac{1}{4}g_1^2g_2^2\left(12m_1m_2C_{\phi 2^2}-\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(9m_1+5m_2)\right)\mathrm{LF}_{3,1,1,-1}[\tilde{\mu},m_1,m_2]-\\
&4C_{\phi 2^2}g_1^2g_2^2\,\mathrm{LF}_{4,1,1,-3}[\tilde{\mu},m_1,m_2]+\frac{1}{2}g_1^2g_2^2\left(-4m_1m_2C_{\phi 2^2}+3\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(m_1+m_2)\right)\\
&\quad\mathrm{LF}_{4,1,1,-2}[\tilde{\mu},m_1,m_2]-4C_{\phi 2^2}g_1^2g_2^2\,\mathrm{LF}_{3,2,1,-3}[\tilde{\mu},m_2,m_1]+\\
&2g_1^2g_2^2\left(-2m_1m_2C_{\phi 2^2}+\tilde{\mu}(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2})(m_1+m_2)\right)\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-\\
&3\,\overline{a_d}^{pr}\,\overline{a_d}^{st}a_d^{pt}a_d^{sr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{st}a_d^{sr}-C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&\frac{3}{2}\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}\left(-\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-\\
&\overline{a_e}^{pr}\,\overline{a_e}^{st}a_e^{pt}a_e^{sr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+\\
&\frac{1}{2}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{st}a_e^{sr}-C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+\\
&\frac{1}{2}\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}\left(-\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-\\
&2C_{\phi 2^2}\overline{a_e}^{pr}\,\overline{a_e}^{st}a_e^{pt}a_e^{sr}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+\\
&2C_{\phi 2^2}\overline{a_e}^{pr}\,\overline{a_e}^{st}a_e^{pt}a_e^{sr}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]-\\
&\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+\\
&\tilde{\mu}\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-\\
&6C_{\phi 2^2}\overline{a_d}^{pr}\,\overline{a_d}^{st}a_d^{pt}a_d^{sr}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+\\
&6C_{\phi 2^2}\overline{a_d}^{pr}\,\overline{a_d}^{st}a_d^{pt}a_d^{sr}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+\\
&6C_{\phi 2^2}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-\\
&3\tilde{\mu}^4\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-\\
&\frac{3}{2}\tilde{\mu}^3\left(C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\,\overline{a_u}^{pt}+\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\right)\\
&\quad\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+6C_{\phi 2^2}\tilde{\mu}^4\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+\\
&\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\,\overline{a_u}^{pt}\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+\\
&\frac{3}{4}\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{sr}\left(-C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\right)\\
&\quad\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\frac{3}{4}\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{sr}\\
&\quad\left(C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}(-\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt})\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-\\
&3\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&3\tilde{\mu}\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+\\
&\frac{3}{2}\tilde{\mu}\Big(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{st}\,\overline{a_d}^{pr}a_d^{sr}a_u^{pt}+\overline{C_{\phi 1\phi 2}}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}+\\
&\qquad C_{\phi 1\phi 2}y_u^{st}\overline{a_d}^{sr}(a_d^{pr}\overline{a_u}^{pt}+\tilde{\mu}^2y_d^{pr}\overline{y_u}^{pt})\Big)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-\\
&\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\,\overline{a_u}^{pt}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+\\
&\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\,\overline{a_u}^{pt}\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+\\
&\frac{3}{4}\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{sr}\left(\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pt}+3C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+\\
&\frac{3}{4}\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{sr}\left(3\overline{C_{\phi 1\phi 2}}\,\overline{y_u}^{pr}a_u^{pt}+C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\\
&3\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\left(C_{\phi 1\phi 2}\overline{a_u}^{pr}+2C_{\phi 2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\\
&3\tilde{\mu}^3\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\left(C_{\phi 1\phi 2}\overline{a_u}^{pr}+2C_{\phi 2^2}\tilde{\mu}\,\overline{y_u}^{pr}\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\frac{3}{2}\tilde{\mu}^3\,\overline{y_u}^{st}\\
&\quad y_u^{sr}\left(C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}(\overline{C_{\phi 1\phi 2}}a_u^{pt}+2C_{\phi 2^2}\tilde{\mu}y_u^{pt})\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-\\
&3\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+\\
&3\,\overline{C_{\phi 1\phi 2}}\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]\Big)
\end{aligned}
$$